

Jonathan Ty

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EDUCATION

University of California, San Diego

La Jolla, CA

B.S. in Computer Science, Cognitive Science Minor

Sept 2023 – June 2027

- GPA: 3.590 Cumulative — 3.837 Major
- *Certification:* Web Developer Bootcamp, AWS Academy Cloud Foundations
- Awarded distinguished \$12,000 Alan Turing Memorial Scholarship for Computer Science leadership and merit.
- Awarded distinguished \$18,000 Regent's Scholarship for academic excellence and merit.

EXPERIENCE

Software Development Intern

June 2025 – Sept 2025

Aimia, multi-agent AI-driven career development platform

San Diego, CA

- Designed, developed, and integrated user-facing components and authentication flows in a Next.js/React application, implementing responsive UI and local and cloud-based storage session management
- State management and custom React hooks (useState, useEffect, useRouter) across 15+ modules
- Designed and integrated RESTful API services using Express.js, Mongoose, and JWT: defined MongoDB schema, created backend profile endpoint, built and connected to MongoDB, implemented secure CRUD operations

Undergraduate Research Assistant

Sept 2024 – May 2025

UC San Diego, Computer Science & Engineering

La Jolla, CA

- Developed an audio-based method for efficient Human-Activity Recognition (HAR) data collection, reducing annotation time by over 70%
- Achieved 81% validation accuracy on audio-labeled HAR data using LSTM models, outperforming both button-based (56%) and WISDM (62%) baselines
- Built data processing scripts in Python to parse 9-axis IMU sensor data, align with audio instructions, and evaluate model accuracy across labeling modalities

PROJECTS

Study Buddy | *React, Bootstrap, FastAPI, MongoDB, Vite, Axios, Google Gemini*

April – April 2025

- Contributed to frontend development in a 4-person team a full-stack web application for users to upload PDFs and generate AI-powered flashcards and quizzes for personalized learning.
- Designed a responsive Bootstrap interface, featuring dynamic components: flashcards, quizzes, and study pack creation.

Fast Food Voting Analysis | *Python, Pandas, BeautifulSoup, Selenium, scikit-learn*

May – June 2025

- Wrangled 12 datasets, normalizing fast-food franchise counts per state using Pandas, NumPy to build machine learning analyses achieving 0.81 average accuracy in predicting 2024 state voting outcomes from per-capita franchise densities
- Automated web scrapers with SSL handling to extract restaurant totals from Fast Food location directories for reproducible feature engineering.

Grocery Web App | *HTML, CSS, JavaScript, Figma, Node.js, Express.js, GitHub*

June – June 2024

- Developed in a 4-person team a full-stack web application with Node.js and Express.js for backend API development, and integrated it with a responsive frontend using HTML, CSS, and JavaScript.
- Composed UI/UX process using Figma, creating wireframes, prototypes, and high-fidelity designs to enhance user experience and ensure intuitive, user-friendly interfaces.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++, HTML/CSS, JavaScript

Developer/Design Tools: Figma, Git, Matplotlib, Numpy, Node.js, Next.js, Pandas, Pytorch, React, SciPy, Scikit-Learn, VS Code

Coursework: *Advanced Data Structures, Algorithm Design, Artificial Intelligence, Machine Learning, Operating Systems, Practicum in Large-Scale Codebases, Recommender Systems & Web Mining, Software Engineering, User Interfaces, Wireless Networks*